

LSTMs and GRUs

DATA 443/643 · Week 2, note 2.3

After CMU 11-785 L14: a cell that can copy, plus input-dependent gates.

Lecture plan

- 1 Why a tanh overwrite cannot remember
- 2 Three LSTM gates and the cell highway
- 3 Copy regime: forget open, input closed
- 4 GRU: two gates, hidden state is memory
- 5 Bidirectional is illegal for next-token LMs

First time: what / why / formula

- What / why. An additive cell path that can copy a value when the forget gate is open and the input gate is closed. GRU is the two-gate cousin. Vanilla RNNs vanish. You need a highway whose gradient is ≈ 1 along the copy.
- Architecture. LSTM: cell c_t , hidden h_t , gates i, f, o . GRU: reset and update, no extra c .
- Formula. $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, $h_t = o_t \odot \tanh(c_t)$. GRU: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$.
- Tradeoffs. + Long-range copy, still sequential. – More parameters than a vanilla RNN; still not parallel like attention. GRU is cheaper, sometimes enough.

How. Copy regime: $f \approx 1, i \approx 0 \Rightarrow c_t \approx c_{t-1}$. Lab 2: LSTM vs RNN on a long copy.

The hidden state is constantly rewritten

Vanilla RNN

Every step: tanh of a new mix. Memory length is an eigenvalue accident.

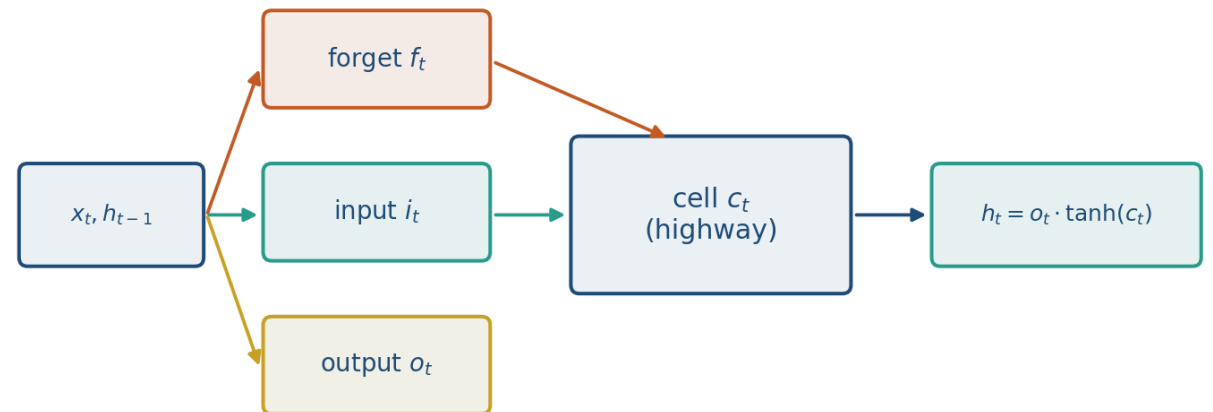
Gated cell

A wire that can copy, plus learned switches that say when to write or erase.

LSTM: a highway plus three gates

- Forget, input, output. Each in $(0, 1)$.
- The cell is mostly multiply and add.
- Hochreiter and Schmidhuber 1997; forget gate Gers 2000.

LSTM: a cell highway plus learned gates



The cell update you will write

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Then $h_t = o_t \odot \tanh(c_t)$.

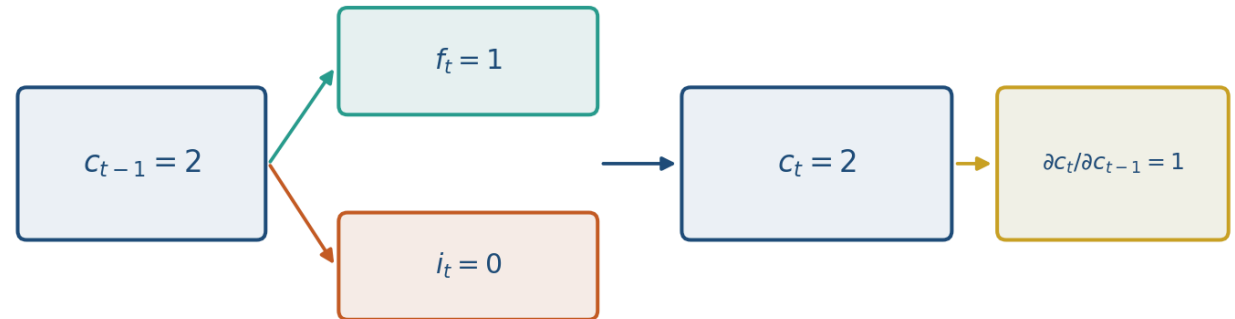
If $f_t \approx 1$ and $i_t \approx 0$, the cell copies.

Along that path,
 $\partial c_t / \partial c_{t-1} = f_t$.

Copy regime

- $c_{t-1} = 2, f = 1, i = 0$ gives $c_t = 2$.
- Ten copies: gradient $1^{10} = 1$.
- Learnable, not guaranteed. About 100 steps, not infinity.

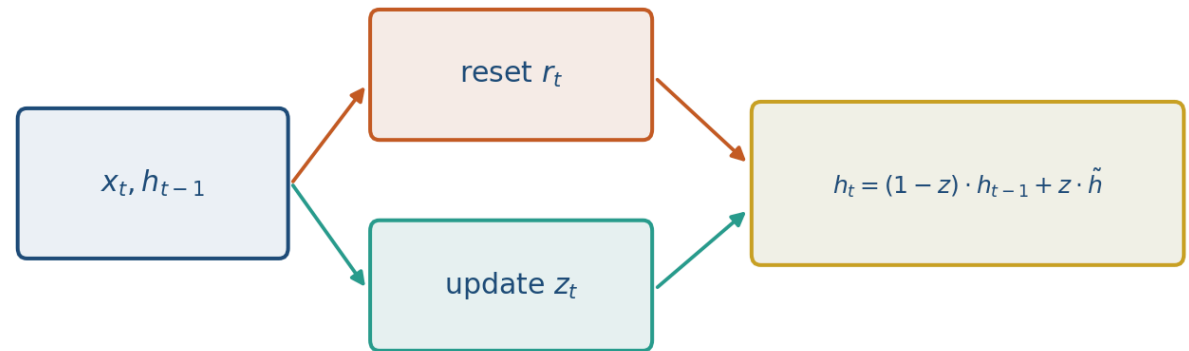
Copy regime: forget open, input closed. The cell is a wire.



GRU: two gates, no extra cell

- Reset and update. The hidden state is the memory.
- $z \approx 0$ copies. $z \approx 1$ replaces.
- Fewer parameters. Still a loop.

GRU: two gates, no separate cell state



When you would still pick a gate

LSTM / GRU

Speech, time series, a small sequential baseline. Lab 2.

Transformer (Week 3)

Long text, GPU parallelism, path length one. The default for this course's projects.

What to take from this lecture

Write it

$c_t = \text{forget} * \text{old} + \text{input} * \text{new}.$

Copy it

$f=1, i=0$ is a wire. Bidirectional
peeks at the future.

Next week

Attention stops using this chain.
Lab 2 is why.